

Feasible Utilization of Machine Learning and Other Technologies to Help Personalized Dermatological Treatment

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Abstract

Artificial intelligence has advanced far and can be applied into the medical field. This can be seen in advancing personalized treatments through genomic analysis or through recommendations based on the patient's profile. In all of this, the physician can utilize these systems to work more closely with the patient and personalize treatment. Alongside this, efficiency and better diagnosis may make AI-based systems a necessity in the future. However, in all of this, the cost must be analyzed as well, whether it be the ethical implications of over dependence, the security risk of patient data, or even the legal implications of when things go wrong. In total, dermatology serves as a good case study to analyze AI's application in the field.

Introduction

The growth of Artificial Intelligence (AI) in recent years has caused quite a stir. Applications ranging from education to games all seem to incorporate AI, and the medical field is not excluded in this. Though not always a useful application, AI seems to carry a promise of higher efficiency with less effort, causing the spread of various technologies and applications that heavily utilize AI. In the medical field, this desire seemed to spark much innovation in hopes of personalized treatments, medical breakthroughs, and physician assistance.

Though an in-depth understanding of machine learning is not needed for this paper, a brief overview would be beneficial. In the state of mid-2025, there have been quite a number of models that have made their debut and made it into the spotlight. Few notable ones include Open AI's Chat-GPT and O1 models, Google's Gemini, Anthropic's Claude, and DeepSeek and its parent company under the same name. These Large Language models, or LLMs, have risen in popularity but are far from the vast amount of machine learning models out there. In general, there are three main types of learning: supervised, unsupervised, and reinforcement learning. Each has its own way of presenting data to the model and allowing the model to change. From these basic types, there has been the rise of deep learning architectures, also known as deep neural networks. These models are built off of layers of nodes, also called neurons, that are layered in an architecturally specific order. Much like biological neurons, these neurons activate for the next layer by computing the activation of prior neurons connected to it, much like spatial summation. Common deep learning models are CNN's (Convolutional Neural Networks) for image analysis, RNNs for recurrent or sequential data, GANs (Generative Neural Networks) for artificial content generation, and even transformers, which constitute the architecture of the NLP and LLMs previously mentioned.

The model creation is broken down into two major phases: training and testing. The training phase is where the model takes in the training data and modifies itself to best figure out the correlations seen in the data, similar to making a general formula after analyzing vast amounts of data. This step is known to be the most intensive in terms of computation and cost. This step is vital to the model, and biases in the data used can easily deter the model, creating a biased model that draws conclusions from erroneous sources. In the testing or implementation phase, the models are deployed. Some are deployed as chatbots, as commonly seen today, and yet others are focused on specific interfaces. In robotics, models are used to autonomously control movement, giving the robot a brain of its own. In direct analysis, live videos can be analyzed for the benefit of the user, such as for safety sensors and cameras on cars.

In this paper, we will take the case study of dermatologists, or physicians who analyze skin diseases. Much of what is seen in dermatology can have very physical (seeable/visual) presentations. Though this ease of access, biopsies (among other testing methods) are a valuable determinant of the disease the patient experiences. The push for less biopsies and more accurate results has risen through the utilization of AI, specifically CNN's. "The massive learning

capacity of AI allows it to recognize subtle differences in lesion features such as size, texture and shades, and far surpasses that of humans” [1]. The question then arises, can you analyze without a biopsy, and how can you be more precise? Can AI help with patient interactions, such as quantizing the desire of the itch, the intensity of the burn, the vague feelings the patient presents with, or even the confusing terminology given as descriptions?

These questions open the door to the world of AI and other technologies, but the implications may not be as expected. The finished models can perform a variety of functions, from surgical applications to assessing live data from the patient. All this can be useful for physicians, perhaps to get a second opinion or for further analysis. Whole processes can be automated, eliminating needless work done by physicians and allowing better and more efficient usage of their time. However, this all may come at a cost, whether it be in the ethical and legal implications, the safety issues raised, or in the actual feasibility of physician usage.

Physician Focus

There are a variety of challenges that plague a physician. In dermatology, these issues are commonly seen in the lack of options available and the quick identification of a skin disease. Much of what improvement can be made to dermatology is not to help the physician become a more efficient machine, but rather to help the patient more. This can be seen in better treatment, more treatment options, and even classifying and quantizing particular data that could give the dermatologist more background into the patient.

As much of dermatology lies in physical presentation and with testing done as needed, the classification of what a patient has is almost an end goal. The following steps after this knowledge come in the form of either treatment or future prevention. Only one (the former) of these are ideal as they give the patient confidence in the hope that a recurring issue will not happen. On the other hand, chronic conditions are most commonly seen in dermatology, and as such, many of the future preventive measures are to help assist the patient in comfort, such as to prevent an itch, or in recovery after an event (irritation, leakage, etc).

For treatment, these major developments can be seen making great strides in precision medicine. In general, precision medicine is focused on “offering personalized and targeted therapies that improve health outcomes and enhance the quality of life” [2]. The idea is that general medicines can be honed to produce a greater affect by considering various variables, such as genetic makeup (primarily), prediction of transport capacity and efficacy, family history, and drug compatibility to name a few [2].

Delving into fields such as pharmacogenetics allows for better understanding of the molecular world and the interactions that shape the functions of the body. The role of a drug therapist is that of creating a molecular cure for a very specific problem. Whether it be through

new drug creation or by modifying a class of drugs, AI has been implemented to help solve these problems aside from just the common brute-force method of trial and error. As discussed by Schork et al., pathology plays a big role in the targeting of treatment, but the “nuanced features of the patient, and the optimal way to correct the pathology given those nuances, are not often clear” [3]. Utilizing AI allows links to be drawn, whether it be in familial genetics and possible therapies associated, or with environmental factors that vary expressivity. “These technologies enable the independent detection and analysis of patterns within the data, providing the ability to predict pharmaceutical properties of drug targets, which is especially beneficial in clinical settings” [2]. An example of the usage given in the report briefly described a study comparing different models for predicting cancer based on gene expression with mutations present. The results showed that the deep learning model identified the major drug sensitive genomic features. “The results confirmed the applicability of drug response prediction models for individual drugs, highlighting the potential of these models in personalized medicine for cancer therapy” [2]. As each genetic makeup varies, it is common to see that some therapies, such as novel cancer therapies, do not always work for everyone. The futuristic view presented by many authors is one in which an model can utilize prior knowledge of successful treatments and their mechanisms to adapt the treatment specifically for a patient. In short, Carini et al. expands that the use of AI “offers enhanced drug effectiveness, improved safety profiles, decreased adverse reactions, expanded treatment choices, and a potential for saving lives” [4].

Alongside pairing genetic makeup and possible effective changes personalized to patients, AI has also made an impact in the world of drug discovery. “The successful approval of a drug requires the concurrent optimization of various properties encompassing pharmacokinetics (PK), pharmacodynamics (PD), and clinical outcomes” [4]. Carini et al. goes on to explain that PK encompasses “absorption, distribution, metabolism, excretion, and toxicity” to the medication, PD focuses on drug-target interactions and drug safety, and clinical outcomes focus on the actual results, including the uncommon or unforeseen side effects [4]. As discussed before, AI can circumvent the trial-and-error approach by focusing on the relations between the drug structure, absorption, metabolism, and toxicity. As pharmacology grows to adopt informatics and AI, the focus has shifted to lower cost and higher efficiency. The majority of current drug discoveries are based on physical lab tests, but with the introduction of AI, virtual spaces and theoretical interactions have now become the focus. Though lab tests are still needed, much of the process can now be optimized for PK and PD.

As another option for treatment, the surgical field is not void of machine learning either. There have been many advances in the field of surgery, most notable of the late-twentieth century being the inclusion of robotics. These tools provided more accuracy for the physicians while also taking out the shortcomings seen in physicians, including tremors or lack of visual images. The dexterity and precise maneuvers robots can perform also serve as big motivators for including them in the surgical setting. Introducing AI into these surgical robots pushes this boundary even further, seeking to reduce basic human error while increasing accuracy and

decreasing time spent in surgery. Having an artificial being in a surgical setting is not meant to perform novel or situation-led procedures, but rather mundane and laborious processes that would help the physician better focus on the main portion of the procedure. Though this is not seen directly in dermatological specific procedures, general maneuvers have been taught to AI models, who, in turn, can replicate the movements necessary to produce the same result. This can be seen in flesh cutting tasks as well as suturing. The STAR, or Smart Tissue Autonomous Robot, is a robot developed by researchers at the University of Maryland, trained to cut much more accurately than even a professional surgeon [5]. This precision goes hand in hand with the experience the robot has due to its training on various types of tissue (muscle, fat, and skin) and what it is cutting out, such as tumors [5]. Though this AI surgical robot was developed with a focus on precise incisions, other teams have focused on the development of robots to learn surgeon movement. An example of this, a team of Stanford and Hopkins researchers trained a da Vinci robot for the task of imitating surgeries [6]. This task included translating the professional movements given by physicians (as seen in videos) into calculated movements for itself. This resulted in a revolutionary system, one that could “perform three fundamental tasks required in surgical procedures: manipulating a needle, lifting body tissue, and suturing” [6]. These current advances of integrating AI into surgery do not directly interact with dermatology. These times, however, are the beginning of the introduction of this new hybrid, so advances will most likely pick up rapidly. “Before this advancement, programming a robot to perform even a simple aspect of a surgery required hand-coding every step” [6]. By simply training and using these AI focused surgical robots, many procedures can be done quickly and with minimal effort.

Easily classifying and quantifying skin diseases would remove much of the ambiguity seen in dermatology. In more specific terms, as dermatology is based on physical presentation of the skin, a diagnosis can be made confidently and quickly through a quick analysis of features such as “presence, morphology, extent, and distribution of lesions[, and] signs of secondary infection (eg, erythema, swelling, warmth, yellow or honey-colored crusting)” [7]. Upon suspicion, it is common for dermatologists to take biopsies of an area of concern for further analysis. This usually is a quick procedure and produces accurate results, so the technique itself does not call for any modification. However, in the face of uncertainty or when time or unprecedented situations arise, it is hard to always have these tests done. As such, a practical application of image analysis via machine learning would be directly in dermatology.

As a fundamental application of AI would be directly in analysis of images, it would be appropriate to discuss the basic advantages that these systems could have. In the wake of Covid-19, the world stood still. This prompted many medical visits to either be canceled or to be taken online. In the case of dermatology, this caused a great number of individuals to present their conditions in their own home with less variables the physician could control. As stated by Mr. Vodrahalli in an interview with HCP, much time can be saved when an AI algorithm can help determine the quality of pictures taken by patients to help the physician better diagnose the issue [8]. This was a common struggle seen during the pandemic, where images may have been of

lower quality, leading to more time spent by both physicians and patients, and possibly even incorrect diagnosis. By utilizing these AI models developed, the quality of the picture is immediately analyzed, leading to more accurate findings. Beyond just the pandemic, this working model shows that using AI can practically help in telemedicine, cutting visit times and saving time for both the physicians and the patient.

A more advanced usage of image analysis could be seen in direct diagnosis. Acting as a second pair of eyes, the models presented to physicians can assist in diagnosis across skin tones and backgrounds, as discussed later. Looking at one of the more complex usages of diagnostic image analysis, we can look at pathology as an example. As a disease focused field, much of the field consists of analyzing a plethora of images taken from tissue samples. As described by aetherAI, an AI focused pathology company, “the traditional practice of pathology has remained largely unchanged over its 150-year history [...] traditional practice is not only manual, but is prone to inaccuracy, error, and inefficiency” [9]. This company provides a model, GigaPixel AI, and shows that high-density images are no longer a problem. Though conventional CNN’s would take a large amount of time and strain to analyze images commonly seen in the field, pathology slides can be streamlined through an “end-to-end weakly supervised method called enhanced streaming CNN (ESCNN)”[9]. This utilizes images with a resolution of 75k x 75k, which is enough pixels to fill up a billboard that covers several city blocks and enough information to slow normal processors. By creating this new model, this downsizes the effort needed to not only train, but also be practically used by pathologists or other physicians who must take in vast quantities of images for analysis [9][10]. Using an example of gastric cancer, which can only be diagnosed via a histological evaluation, the tool can easily be used to visualize which parts of the image contain lymph nodes and gives a heat map, an attention map, and a prediction map [10]. The model can differentiate between macrometastasis, micrometastasis, and isolated tumor cells (ITCs) faster than other methods. Though not the main purpose of this paper, the novel approach taken in training this model contained a high throughput and low memory usage for both training and testing, both of which are vital in understanding the costs of these systems (discussed later). This tool fits well into the practice of physicians as they can control and modify the images, even correcting the boundaries of what the model detected as a lymph node to help get higher accuracy. When tested with trained pathologists, the pathologists performed better than the model over most images [10]. For ambiguous images, however, the model helped gain a higher accuracy for those who used the tool on a second review in a comparison to those who did not use the tool [10]. Though not in the direct scope of dermatology, these results can be directly applied to the field as this is an advanced example of the possible tools that can be used to help physicians come to a diagnosis confidently and quickly.

Patient Focus

AI and other emerging technologies have the capability to not only affect how physicians work, but also how patients view medicine. As a physician is more than a robot, a physician's role is to create a therapeutic relationship with the patient, taking into account their background, and catering a treatment plan for them. As an extension of personalized medicine, if realistic AI systems are to be used, AI must be able to take into account the patient's background and provide good suggestions to the physicians and the patient. This extension could even be provided in specific technological applications catered to the life of the patient.

Taking into account a background is more than just race and age, but also the patient's culture, religion, food and diet restrictions/preferences, the place they live, the job they have, and the relationships they are in. Possible stressors need to be discussed, and family history needs to be shown to help create a full profile of a patient. Human physicians naturally do this through natural conversation or pointed discussions. Though physicians can have all this information, this does not always make the physician all-knowing or provide the physician with a full view of the patient. Providing practical lifestyle changes catered to the profile of the patient helps the patient adjust and may increase the adoption and effectiveness of a treatment plan. The similarity of treatment to their normal life reduces the resistance to treatments, whether it be in stigmas associated with treatment or cost concerns. This can be seen in AI help to create culture or religion specific diets, creating personalized fitness goals based on time available, and even incorporating mindfulness techniques that help the patient realize their stressors. Beyond this, catering technologies and advances in research directly into the life of the patient incentivizes them to actively participate in their treatment.

Specific to dermatology, surgery and medications may not be the only options, or the most effective. Providing options for patients pushes technology to be developed to cater to each patient. This has led to great advances in personal health technologies, as seen in Apple Watch's, Fitbits, and many more compact technological devices. These options provide physicians and patients with data but lack an application for treatment or strict management. A great point brought up by Dr. Xu, "[if you think about it], why do we have electronics strapped to your wrist? It's not because that's where the data is, it's where it's convenient to put electronics; but if you have skin-like soft-flexible sensors that you can put anywhere, there's areas of rich information" [8]. He then goes on to explain the importance of an "itch" in the context of the patient suffering from, as an example, atopic dermatitis. Quantifying and classifying the data, figuring out the triggers of an itch and preventing it, perhaps without even medication, would be the revolution needed in physical based medical technologies. This is the focus and basis of therapeutic textiles and smart textiles.

Therapeutic textiles focus on the relationship that exists between the skin microbiota, common skin diseases, and the clothes we wear. "While the topical use of steroids and antimicrobials in patients with skin dysbiosis (imbalance of the skin microbiota) has become a significant health concern due to its prolonged usage and long-term side effects, textile materials have gained particular interest as a non-pharmacological approach due to its direct contact and

ability to induce specific changes in the skin microbiota with no side effects” [11]. The reasoning is that the microbiota for patients suffering from dermatological issues have a varied or suppressed microbiota. The long-term treatment of such conditions is known to create issues that may perpetuate the issues more over longer periods of time. By utilizing clothing, however, the microbiota can be restored or enhanced to protect and function in a more normative fashion. “The microorganisms living in clothing textiles have been a major target for odour control and disease prevention. The interaction between the wearer and clothing is an opportunity for skin bacteria to attach to the textile surface, which can lead to the growth of certain strains” [12]. The results of these investigations show that there is a correlation between the clothes we wear and how this affects our microbiota. Advances in this field have been in the form of chemicals to add in the making of the fabric, different types of materials, or even various treatments of the fabrics. As of importance to athletic textile manufactures, research has focused more on preventing the growth of the microbiota as this is correlated with “permastink” fabrics and the increase in foul odor. Of importance to dermatology, however, the fabric could actually be the method that is used to treat and promote a better skin biome. As discussed by [11], the analysis of the dysregulation of the microbiota should actually be the leading factor for the patient in treatment of the disease. “Therefore, for the control, treatment and management of skin dysbiosis, it is necessary to specifically target the skin microbiota imbalance and restore the commensal species. This property is considered a key feature to modulate the host inflammatory response and improve the symptoms of the disease by supporting a healthy skin environment” [11].

On the other side of the textile approach to treatment is the direct integration of technology into the fabric. As presented by the introduction of 3DKnITS and SeamFit, new fabric is now under development that can incorporate AI to recognize movement. This is vital to the concept proposed by Dr. Xu as it would help classify movements and actions that may propel the symptoms seen in patients suffering from dermatological diseases. Both 3DKnits and SeamFit approach integration in different ways. As described by Wicaksono et. al, “[t]here are two common methods of pressure sensing in electronic textiles: capacitive and resistive-based” [13]. Both opt for the different options, leading to varied implementations, though they intersect in one vital technique. When analyzing the structure of 3DKnits, the conductive nylon is integrated directly onto polyester/spandex, allowing for stretching of the fabric while maintaining and mapping changes in resistance [13]. This means that the fabric must be directly in contact with the skin to accurately read the changes, unlike in SeamFit. Focusing on multi-use, variable sizing, and loose fitting clothing, SeamFit integrates electrodes into commonly seen seams in T-shirts [14]. As the shape of the clothing changes, the conduction changes as well. These changes are then recorded and used to deduce the workout being done. Resistance in 3DKnits, utilizing piezo-resistive textiles, offers pressure data as well as resistance data, all placed within the confines of a polyester fabric. Both applications, however, just collect raw data and attempt to output the correct motion or exercise. This is assisted by their own trained AI models, one using a CNN based model and the other a CNN-LSTM Encoder model [13][14]. As seen, these models alongside these technologies provide in-depth information about position and

movement, all which can be vital to creating personalized treatments. As these applications were both created for exercise and movement, the utilization of such technologies in dermatology can almost be directly applied. Understanding certain factors, such as environmental and positional awareness, can help notify the patient and the physician of certain unhealthy trends, allowing practical changes to be made in the patient's life to avoid such factors and promote health. Though no current technologies are proposed for embedded smart textiles with implicit dysbiosis detection, the trend of these technologies show that inclusion of these factors could be the next step. The combination of the material science in encouraging good microbial dermatological growth alongside the technological innovations to help track and perhaps even assist creating a movement profile show that treatment can go beyond just the surgical or strict medication approaches.

AI Assistant

The word most associated with artificial intelligence would be “efficiency”. As physicians struggle under the immense weight and responsibility of their job, many have tried to alleviate physician’s struggles through a variety of methods. Integration of AI into practice has been proposed, but the practicality of usage has yet to be seen at a large scale. A common approach that has gained traction would be integration into physician’s EHS workflow. Though this is a vital part of a physician's job, are there alternative implementations that would lead to higher efficiency?

One of the common complaints of physicians is that they are overworked. It is now more common for physicians to bring their work home, whether it be in the form of writing patient charts, documentation, or approvals. This transition has been gradual, but with the change has brought an increase in overtime work, increased stress, and less work-life balance. AI has seen many phases in application, one of the most prevalent and useful being as an assistant. Note writing up to new standards can be learned and implemented with minimal effort from the physician. Common tasks that take excessive time can be put through a normal ML workflow, thereby reducing the burden on the physician and increasing efficiency. This, in its purest form, is the definition of an AI assistant, one that fully knows the requirements and preferences of the physician it is assisting all in hopes of decreasing the strain felt by the physician.

As a case study in dermatology, the machine assistant can be utilized for direct image analysis. As stated by Groh et al., “the effectiveness of physician–machine partnerships remains an open question, in part because physicians and algorithms are both susceptible to systematic errors, especially for diagnosis of underrepresented populations” [15]. As brought up by the paper, it is well documented that textbooks and other learning resources lack a diversity in skin tones, leading to many common cases where physicians could not correctly give a diagnosis as fast or as accurately in comparison to when they are presented in a previously studied skin tone

[15]. Focusing on variances in the data can provide useful information to AI based systems to see common features beyond just the skin tone.

Utilizing a tele dermatology context, or solely depending on image presentation, a study was designed to allow physicians of a few categories to present their diagnosis. The categories of physicians included various quantities of dermatologists, dermatology residents, primary care physicians, and other physicians. The physicians were allowed three initial separate diagnosis and were provided with skin disease images, including diseases such as atopic dermatitis, secondary syphilis, and dermatomyositis, all presented in a variety of skin tones. Overall, the average accuracy of the physician's diagnosis was four percentage points lower in comparison of skin diseases presented on light skin versus dark skin, with each disease and each group having a slight variance. The study also allowed physicians to utilize deep learning systems, or DLS, assistance following the initial diagnosis. The physicians were randomly assigned to one of two versions of the DLS. One had an accuracy of 84% and was labelled the 'treatment DLS' [15]. The other functioned as a 'control DLS' as it had an accuracy of 47% [15]. Alongside this, each DLS had its own interface, the control containing three buttons, and the more accurate containing the same three but in the reverse order: "'Update my top prediction with [disease]'", 'Update my differential to include [disease]' and 'Keep my differential'"(initial order)[15]. With just the control DLS assistance, dermatologists saw an increase in accuracy of 33%, and primary care physicians an increase of 69%, both in relation to their leading diagnoses. In cases where the DLS made an error, the accuracy of prior groups only suffered 1.2 percentage points, meaning that the physicians overrode the DLS decision more often in cases of erroneous suggestions [15]. Unsurprisingly, when analyzing the treatment DLS, the average accuracy jumped higher than the control DLS. "The goal of this experimental set-up is not to assess the DLSs themselves, but rather to understand the impact of the DLSs on human decision-making" [15]. In reality, these results demonstrate the weight AI systems have in practice as they drastically impact medicine, even in the minor details. When analyzing a common struggle seen among dermatologists, that of determining a diagnosis based on varied presentation in skin color, it is easy to see the value that AI can bring as it helps to surface knowledge the physician may have, though not activated due to abnormal or unfamiliar circumstances. Though the usefulness of this system would help the physician become more accurate in teledermatology, factors such as patient history and other presentable variables may actually help more. The inclusion of AI is not perfect, but this study shows the effect a knowledgeable assistant can have to help draw out knowledge and hone the physician's diagnosis.

However, introducing more tools and technologies is a cautionary tale. The role of an assistant is to take some of the burden, not to add additional burdens. Much of what may be seen today would be considered unreliable or may add additional work for the physician. This is due to the increase in technological use required by modern physicians. A study focusing on potential technological stressors and their affect concluded that a negative correlation was seen between years in practice as a physician and both technology use and burnout [16]. "Our analysis showed

similar results in both cases, which suggests that technologies other than EHR, such as the ones we looked at — social media, texting, email, and information look up tools—should also be given consideration when researching associations between technology and physician burnout” [16]. This shows that the role of an AI assistant would have more of an application and advantage in a physician’s life beyond just an EHR extension. To account for and assist in the full life of the physician would further help distinguish between the work and the life of the physician. This, in turn, may lead to better care for the patient and more fulfillment for the physician. Interestingly, the correlation suggests a few possibilities. One such possibility would be that younger physicians have more burnout due to increased technology usage, as opposed to older physicians having less burnout due to decreased technology usage. Another possibility would be the relationship between age and burnout rates, which could be associated with the lack of dependency from family later in life, or even through gained experience to creating a good work-life balance. Research is needed into which of these (or more) factors play a role in burnout. These initial findings, however, underscore the importance of physician-focused tools.

Ethical, Legal, and Security Implications

The development of AI has spiked over the past few years, bringing great advancement in a variety of fields. Though this progress is notable, there are a few downsides associated, such as the environmental cost, the ethical and legal implications, security risks associated, and the consideration if there is any benefit for the physician.

As development increases, large investments have been made in more powerful energy sources. This is due to the general cost of running and creating these models. As the world normalizes the use of AI, the demand for it puts a certain strain on key resources. Among the major resources discussed in the event “The Hidden Cost of AI: Unpacking Its Energy and Water Footprint”, the general environment, the electricity grid, and water usage are of key concern. The discussion first pointed to the training stage of the model as one that requires a tremendous amount of resources, especially since it is computationally heavy. “One speaker pointed out that, for some models, inference’s cost now outweighs the training cost after 50 million usages” [17]. The speaker continued by discussing how companies were not incentivized to optimize AI’s energy usage, so financial gain was prioritized over environmental responsibility or sustainability [17]. Secondly, the discussion turned to electricity and electricity grids. A report given by IEA outlining energy and AI gave predictions on the energy demand on AI and AI’s practical usage to solve these and other climate issues [18]. When analyzed by other researchers, the findings were said to be vague and underestimated [19]. Either way, their analysis of the report detailed a jump of electricity usage at datacenters from approximately 415 TWh in 2024 to 945 TWh forecasted for 2030. Though not all AI related, a large portion was attributed to it. Speakers at the climate event expressed that though “AI contributes to growing energy demand, it is not the largest driver of global electricity consumption”[17]. Thirdly, the use of water was discussed in its

relation to AI. Datacenters that house the servers necessary to run these large models are in constant need of cooling. The best source of such is water, especially fresh water, all of which is obtained locally from the nearest sources to the datacenters. “This not only affects the environment but also poses risks to local communities that depend on this water for drinking and agricultural use” [17]. The lack of standardized measurements was also pointed out as a key factor in the hidden footprint water consumption has [17]. Upon discussing the price of water, environmental concerns, and electricity usage, it became apparent that companies need to be regulated and checked, not only on costs and consumption, but on the ethical principles and sustainability of the resources used to create these models.

The second major downside associated with AI integrated into medicine would be the ethical implications and security risks. “Potential biases, interpretability issues, and ethical considerations are significant in the context of AI applications in healthcare, including pharmacogenomics” [2]. As described by Taherdoost et. al, biases in the data can easily deter the model. This makes it unclear whether the model is accurate according to beneficial and actual standards, or whether it is accurately drawing false correlations. In practice, this could lead physicians to make incorrect diagnoses or false drug creations when analyzing a common yet biased sequence in a genome. They later go on to mention the challenges of making an AI model that was transparent in an attempt to combat biases [2]. In the case of genomic data, biases can easily be introduced due to a low quantity of a specific case, such as a minority group or a rare genetic disorder. Making complex models transparent and interpretable means breaking the ‘black box’ commonly associated with AI, a puzzle that goes far beyond medicine, yet is crucial to the field. “The integration of AI in healthcare, including pharmacogenomics, raises ethical concerns related to data quality, privacy, bias, and the potential for disparate impact on underrepresented groups” [2]. As data is needed to train the model, there must be an ethical standard for sharing private patient data, and a large quantity of it. Alongside this, to cover “edge cases”, minority and underrepresented groups must also participate to reduce the possible bias created by the model. Even so, as described by the title of these groups, the quantity of data may be low and will create false predictions by the models, which may be of detriment to the patients.

Yet another major ethical issue is the overdependence on these systems. “As AI evolves, human doctors have a risk of being highly dependent on AI-assisted diagnostic systems and losing their enthusiasm for learning and self-improvement” [1]. Although much of the speculation of AI dominance is informed and reinforced by movies such as Terminator, the overdependence on these systems can cause a deficit in knowledge and can lead to poorer autonomous decision making, which is vital for physicians [20]. As such, the systems proposed should be utilized as a tool, not to replace the physician or other vital members of the healthcare profession. The judgment and education of a physician is vital to their role in a patient's life and treatment, so the usage of AI in medicine should be to help decrease the amount of time needed for a physician to diagnose a patient, but not to make the physician make unwise and hasty decisions. Alongside this, the physician has the right to overrule the AI's suggestions in treatment

due to their experience and relationship with the patient, both of which heavily influence the course of treatment.

The legal issues essentially come down to when these systems fail. “The provision of ‘decision support’ information to physicians and health care workers – especially that derived from AI-based analyses – opens up a number of thorny ethical issues [] such as who to blame if the use of the decision support leads to poorer outcomes” [3]. As others recommend, the liability falls on all parties, the developers as well as the healthcare team. In systems that simply help recommend or show the physician knowledge, the physician is ‘to blame’ as the knowledge gained as a physician should guide the diagnosis and treatment, not an overdependence on the tools at hand as explained before. This does not, however, put the physician at fault when treatment does not work as planned or other such cases due to the fact that physicians can make mistakes and are error prone. The point of these systems is to support the physician and reduce the amount of errors. On the other hand, functional systems, such as robotics and smart textiles, lie in the developers’ hands. These systems are made with a patient-centered approach, so a recommendation by the physician does not imply their full knowledge of the system, but rather the potential benefits and drawbacks.

Finally, the biggest question in all of this would be the beneficiality in the life of a physician. As discussed previously, the role of an AI assistant would be to reduce the administrative burden beyond just an EHR [16]. Studies need to ensure that the use of AI in practice would not hinder the physician’s normal workflow but would rather enhance it. This key piece of information would help decide the implementation and factors needed for physicians prior to enforcing all physicians to use such tools.

Conclusion

In conclusion, AI alongside other growing technologies have soon become the cornerstone of innovation, especially in the world of medicine. It has long been known that physicians struggle with the tremendous amount of work required by them. The hopes set by AI in other fields show that it can be used to help speed up work by reducing the amount of menial tasks needed to be done. This hope carries on in medicine, to reduce the amount that physicians are required to do. These models can take in large quantities of information and present the physician with a more holistic understanding of a patient, allowing the physician and patient to make more informed decisions. This is what attracts physicians and so many others to artificial intelligence as it provides us a means to escape the tasks we see as complex or rote and allow the human element to return to the field of medicine.

The trend of these technologies shows that these AI models can only get better, and can be integrated more easily. Though the material costs of creating these models are immense, bringing awareness to the subject helps to focus development on creating smaller and more

focused models. As we progress onward, the expectation of these technologies is not to make more general models, but rather to create a few heavily specialized ones that can be harnessed as tools for healthcare providers.

These assistants should be able to model what is needed in the life of a physician to help reduce their burdens and that of other healthcare team members. In the discussion of dermatology as a specific example, this could take the form of a scribe that can note much more than just words, but also expressions and help the physician get a holistic view of the patient. Alongside this, these models can act as a second pair of eyes. These assistants serve by visualizing samples, assessing, diagnosing, and even sometimes drawing out knowledge possessed yet suppressed by the physician. A practical example of a current technology was seen in GigaPixel, showing the physician predicted areas and allowing the physician to edit as needed and make the final diagnosis.

For dermatological surgical procedures, AI driven surgery robots can help deliver the pre and post-surgical aspects with more precision and less time, allowing the physician to focus more on the vital aspects of the procedure. In dermatological treatments, AI models can be used to help cater treatments specific to patients, whether it be through pharmacogenetics and personalized medicine, or simply through recommendations on clothing and practical life changes. An extension of this would be the smart textiles used to treat dermatological disfunctions either by recognizing dysbiosis or by helping the patient recognize their actions. AI has far reaching implications and can be integrated into almost every aspect of treatment, all in hopes of increasing treatment efficacy as well as physician efficiency.

This efficiency, however, should not be at the cost of physician autonomy. Regulation and input from physicians are needed to better assess and understand the complexities of how AI could negatively affect physicians. This can be seen as another enforced tool that physicians must use, causing the physicians to have more technostress rather than an increase in efficiency. On the other hand, catering the tools to specific practices and guiding development by the needs and practical time and usage of the physician may create a better understanding of the systems and may integrate more fluidly. Regulations must also be kept as patient data is vital to training these models and could be a source of attack. Alongside the safety considerations, we must also consider the ethical issue of how physicians may create an over dependency on such tools. As AI has been known to replace certain aspects of jobs, the brain and human components of the physician should never be replaced or dulled in this process, but rather enhanced.

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Recommended Readings

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